

Objectives

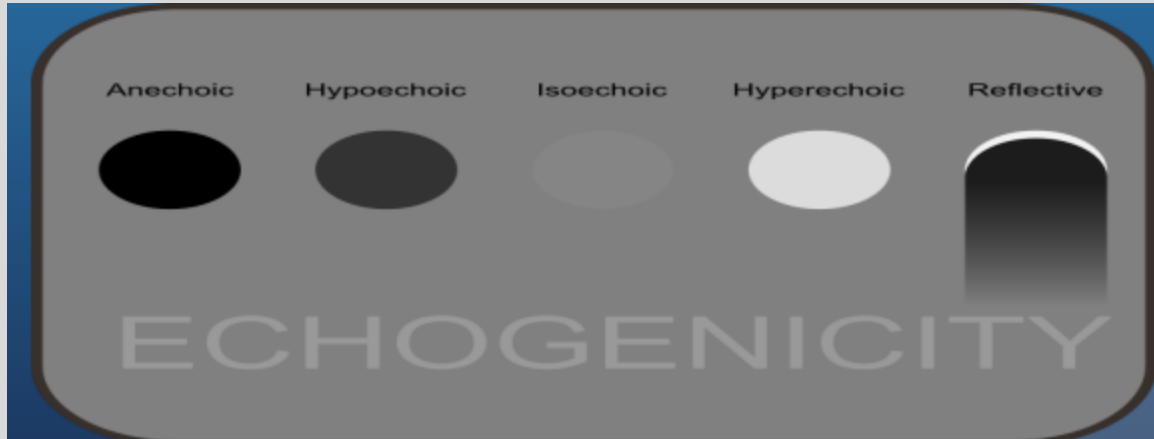
- Student will understand and describe the appearances and differences of tendons, muscles nerves, bone and bursa as well as Anisotropy
- Student will be able to describe common pathologies such as neuropathy and tendinosis
- Student will be able describe indications and limitations of musculoskeletal ultrasound
- Recognize various pathologies presented and compare with normal anatomy

Sonography advantages

- Readily available and relatively inexpensive
 - often performed as targeted exam
 - can often be preformed to accommodate patient positioning
- Real-time
 - performance of provocative maneuvers, ie. demonstrate tendon subluxation
 - direct visualization during needle placement ☐ Non-ionizing radiation
 - useful in children and during pregnancy
 - Can be used in patients who are not MR compatible, pacemakers or claustrophobia

*Gray scale terms

- Echogenicity- brightness on the monitor
 - Hyperechoic- increased echogenicity- brighter
 - Isoechoic- similar echogenicity to background tissue
 - Hypoechoic- decreased echogenicity- darker
- Anechoic- black (simple fluid)
- Shadowing (reflective)- the beam is reflected back (calcium/bone)



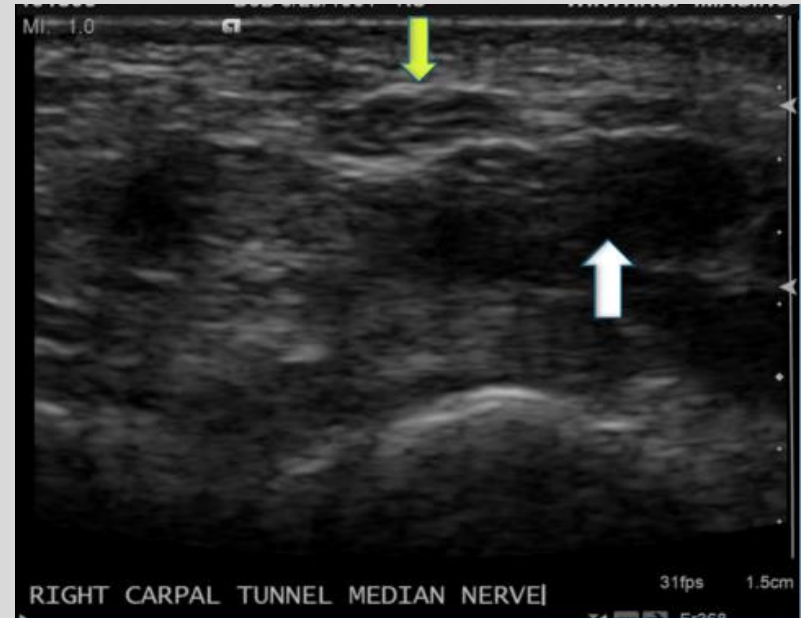
*Nerves



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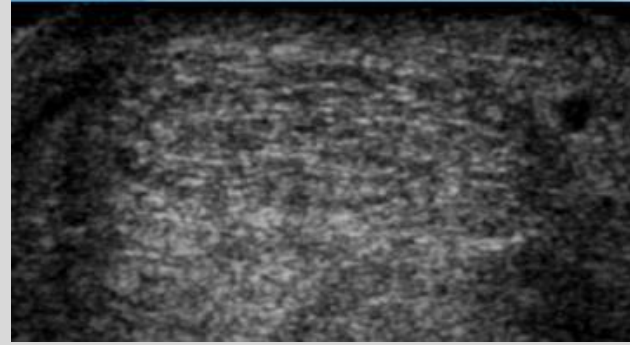
- Round hypoechoic well-defined focus with a homogeneous hyperechoic rim (yellow arrow)
 - Internal hypoechoic foci = nerve fascicles
 - Hyperechoic rim = epineurium.
 - Transverse image- *Honeycomb image
- Compared to tendons, nerves show differences in echotexture, and tendons tend to be more hyperechoic.
- When in doubt dynamic maneuvers of the fingers or joints are helpful in differentiating between mobile tendons and comparatively static nerves during scanning.
- Do NOT exhibit anisotropy (white arrow tendons



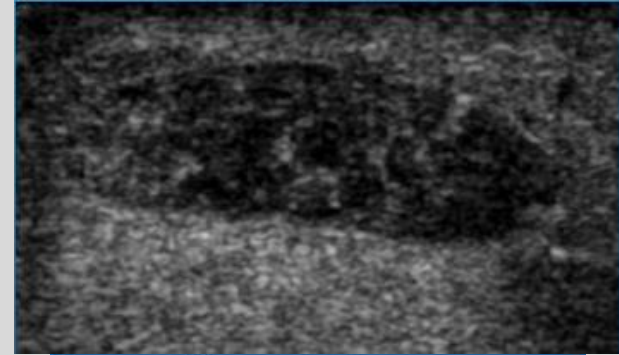
*Anisotropy

- Artifact resulting in apparent decreased echogenicity of the tendon or ligament when the probe is NOT 90 degrees to the ultrasound beam • The reason for this is that the probe will only receive the reflected sound if the beam strikes the surface at a right angle
- Can be mistaken for tendinosis
- Common at tendon insertions and when tendons curve

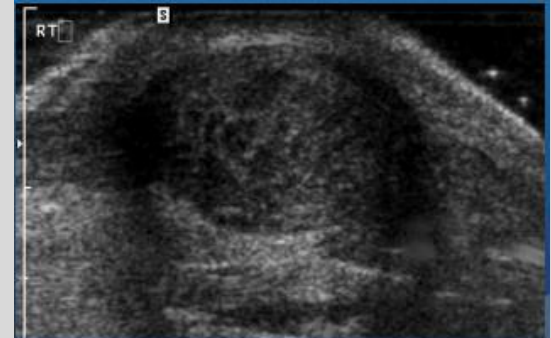
Beam
perpendicular to
tendon axis



Beam insonates tendon at
an angle other than 90
degrees = Anisotropy



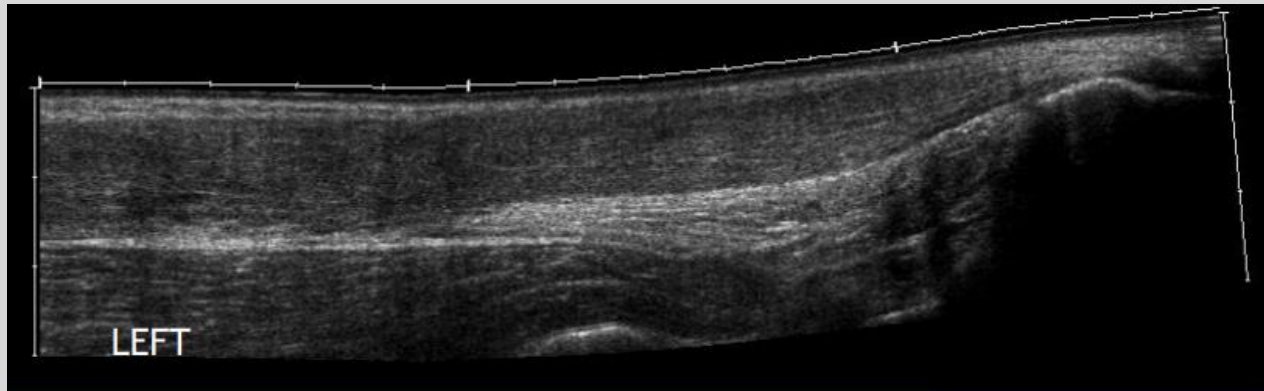
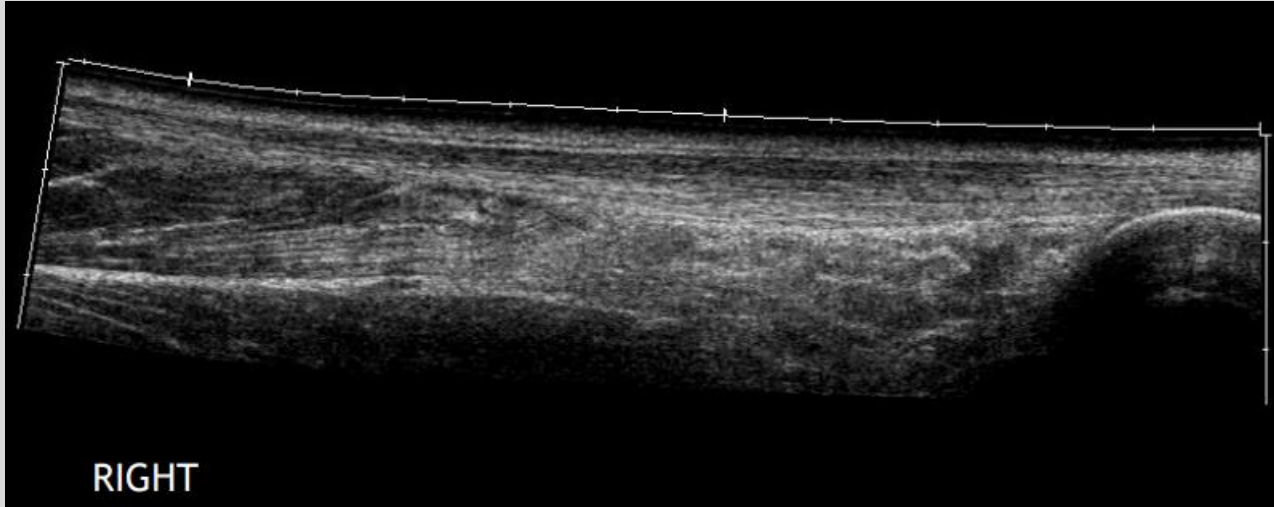
Tendinosis



Achilles Tendon- Which is normal?

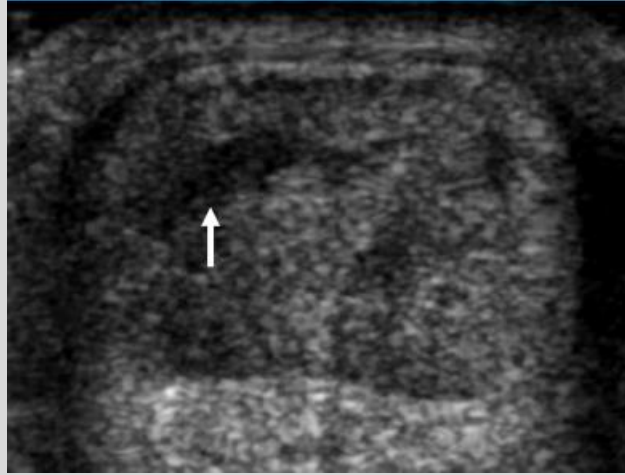
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Tendon tears

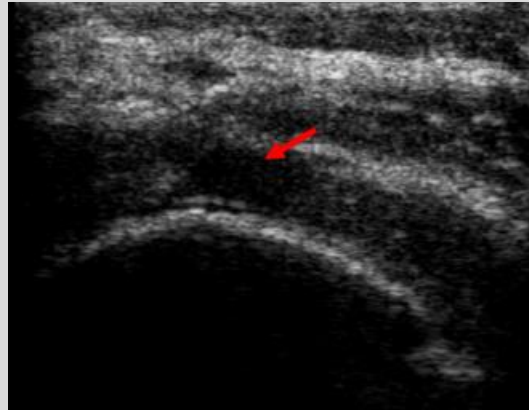
- Partial thickness
 - Intr substance (i.e., longitudinal splits)
 - Does NOT transverse through the entire tendon substance
- Full thickness
 - Extends full thickness of tendon
 - Fluid filled gap



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Transverse view of Achille's tendon demonstrating tendinosis and an interstitial split tear.



Transverse view of supraspinatus tendon demonstrating a full thickness tear anteriorly

*Neuropathy

- Sonography entrapments, trauma and tumors
- Dynamic imaging for nerve subluxations ie ulnar nerve
- Neuropathy- increase size and decreased echogenicity when inflamed, often may see a caliber change (Arrow)



Case 1

- 18 year old female presenting with shoulder pain after attempting cross-fit shoulder press.
- PMH
 - Heart disease with pacemaker
- Limited exam due to pain
- Which imaging test is the next appropriate?
 - X-rays
 - CT
 - MRI
 - Ultrasound

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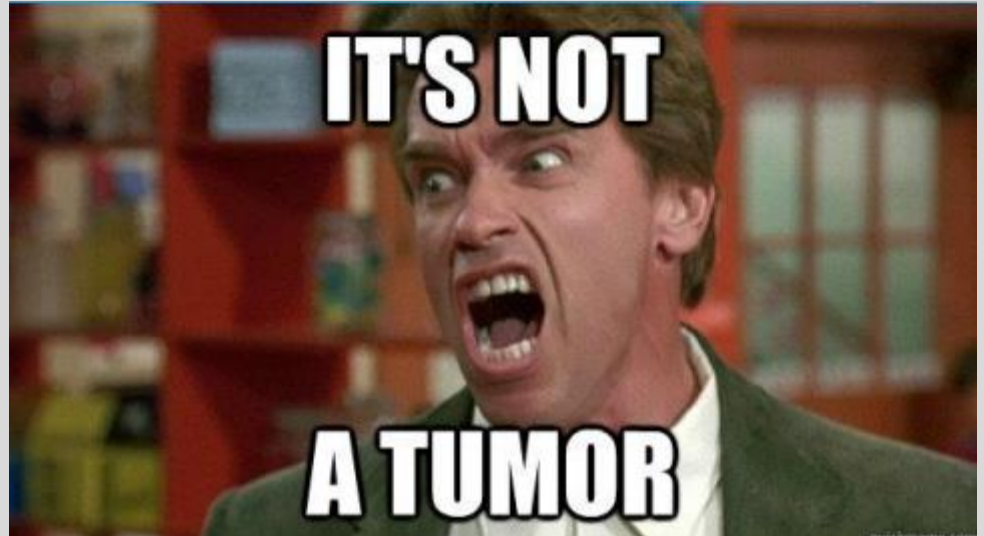


Case 1

- DDX:
 - Rotator cuff tear
 - Labral tear
 - Fracture?
 - Calcific tendinitis
 - Tumor
 - ... its not a tumor

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What next?

- CT-
 - Radiation
 - Good for fractures
 - CT arthrogram for rotator cuff and labral tears
 - Routine CT NOT useful for imaging the rotator cuff or labrum
- MRI
 - Gold standard for rotator cuff and labral pathology
 - NOT in this case... why?
- Ultrasound
 - Ideal in this patient
 - No radiation
 - Dynamic testing
 - NOT good for osseous pathology
 - That's well not entirely true

What can/should we look at with ultrasound?

- Rotator cuff
 - What are they?
- Biceps tendon
- AC joint
- Glenohumeral joint
- Nerves

Case 3

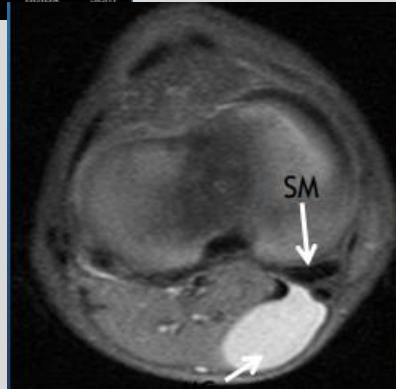
- 45 year old female with increased pain and swelling behind the right knee
- No PMHx
- Refuses to take x-rays because of radiation



- Ddx
 - ** DVT
 - Mass
 - Malignant
 - Sacroma
 - Lipoma
 - Nerve sheath tumor
 - Cyst

Popliteal “Bakers” Cyst

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Bursa/neck of “cyst” between the semimembranosus tendon medially and medial head gastrocnemius laterally adjacent to posterior aspect of the medial femoral condyle.

Case 4

- 35 year old weekend warrior with posterior ankle/calf pain and swelling.



Ddx

- DVT**
- Achilles tendon pathology
- Plantaris tendon tear
- Muscle strain/herniation
- Plantar fascia

Conclusions

- MSK US – dynamic exam with history taking ☐
Should be performed by the radiologist or by well trained technologist
- Anatomic location, relationship to adjacent structures, and history are usually more helpful in rendering a diagnosis, rather than standard imaging features